

PORTFOLIO · 2026

# Nick Lian

I take ideas to shipped software, alone, with AI agents — and set the rules that keep them from breaking.

**UBC Sauder, Class of 2026** · Finance + Business Analytics

Open to anywhere in Canada

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SIX MONTHS, ONE PERSON

# Four kinds of things I built

The numbers are output, not users.

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38

projects

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## Delivered to others

- Reference-letter system  
in use by a professor
- SQL teaching explorer  
no-code tool for a professor

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801

logged updates

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## Public

- Scholar Outflow Lab  
13 countries, updates daily
- Earnings-call sentiment  
tone → residual signal
- Ghost-job trading signal  
310K postings

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25

apps on one private database

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## Used every day

- 25-app personal operating  
system

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8

public repos

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## Runs on its own

- Multi-agent content pipeline  
topic to release pack
- 8 scheduled jobs
- Guardrails and sentinels  
machine blocks dangerous  
operations

## FOUR WORKS

# What I built, and the decision inside each

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### WORK 1 · DELIVERED TO A PROFESSOR

#### Reference-letter system

Roster in, students screened by grade and attendance, documents uploaded, emails out. Shipped as a web app in a day; rebuilt as a desktop app after reading the privacy law, so student data stays on the professor's machine.

3 months · 55 iterations · Python / Flask / SQLite / OCR

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**THE POINT** Nobody asked. I read the law and changed the architecture.

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### WORK 3 · RUNS ON ITS OWN

#### Multi-agent content pipeline

Topic to release pack across several content channels. Agents exchange files, never chat; resumable state machine, bounded retries, a QC loop that sends failures back, and a human always publishes.

32 modules · 20 skills · usage ledger reconciled to the platform bill

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**THE POINT** My own ledger covered 1% of the real bill. The bill became the source of truth.

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### WORK 2 · USED EVERY DAY

#### A 25-app personal operating system

Single-file apps with no build step and one shared convention set, all on one private database. The website reads only data marked public and has zero access to the rest.

25 apps · 10 with built-in AI · count gate before every deploy

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**THE POINT** Permission boundaries fixed in the architecture, not in habits.

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### WORK 4 · KNOWLEDGE

#### Structured knowledge bases built with AI

Thousands of entries across several private libraries. AI does the entry work; every entry carries a permanent ID and a source; reuse is by pointer, never by copy.

Verbatim first · no lossy simplification · checks before and after every change

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**THE POINT** The hard part is the acceptance discipline, not the AI.

# Six public projects

Each one is a complete, documented tool.

## Query Mirror

No-code SQL teaching tool for a professor: click tables, filters and groups, the SQL appears live. Deterministic by design, no LLM.

Single-file HTML + vanilla JS  
[NickkkLian/SQL-Viz-for-Edu](#)

## Scholar Outflow Lab

Where researchers from 13 countries end up, from OpenAlex open data. Ranked by Wilson lower bound so small samples sink. Updates daily.

Python stdlib scraper · GitHub Pages · launchd  
[NickkkLian/Scholar-Outflow-Lab](#)

## CallDelta

Management framing vs analyst pushback, by topic, across quarters. Does tone explain the residual move net of the EPS surprise?

Python · tool use + Pydantic · React dashboard  
[NickkkLian/Earnings-Call-Sentiment-Analyser](#)

## Vacancy Signal

Ghost postings detected in 310K listings, turned into a cross-sectional signal. Walk-forward backtest; +3.7pp/yr vs SPY, 2019–2026. Team of three.

XGBoost · GMM · SHAP · Revelio data  
[NickkkLian/Ghost-Job-Detection-And-Trading-Signal](#)

## Clipbind

Saved Xiaohongshu posts and Bilibili videos in one library; Claude merges selected notes into one writeup. Data stays in your browser and your own repo.

Single-file app · IndexedDB · MIT · live demo  
[NickkkLian/content-organizer](#)

## ApplyLedger

A multi-region job-search tracker in one HTML file: prompt generators, batch resume tailoring, and the GitHub Contents API as its backend.

React 18 in one HTML file · GitHub Contents API  
[NickkkLian/Job-Tracker](#)

# Ship first, then let machines keep it honest

Every rule comes from a real incident. I don't hand-write code: I define the problem, set the constraints, verify the result, and own the result.

## Keeping systems reliable

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- A pre-tool hook blocks dangerous operations  
force push, mass deletion, pushing while behind · 332 lines · ask, never deny
- Sentinels raise alarms  
database shrinks, copies drift, sessions collide
- The reviewer is never the doer  
an audit agent judges evidence only
- Three backup tiers, restores actually rehearsed

## What I can do

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- Take an idea to production alone
- Set the rules that keep systems reliable
- Python backends, single-file frontends, agent orchestration  
13–15 AI sessions in parallel, one repo each
- Product judgment: what is worth building, what to cut
- Business grounding: valuation, financials, backtesting  
bank valuation work · CFA Level I